

AS Level Mathematics — Pure Mathematics 1 (P1)

Practice Paper 10

Time allowed: **1 hour 50 minutes**

Instructions to candidates

- Answer **all** questions. Write your answers on paper, showing all necessary working. Marks may be lost for insufficient working.
- If a numerical answer is not exact, give it correct to **3 significant figures**; give angles in degrees correct to **1 decimal place**, unless a different level of accuracy is specified in the question.
- A scientific calculator may be used. Do **not** use a graphical or symbolic calculator.
- The number of marks is given in brackets [] at the end of each question or part question. The total mark for this paper is **75**.
- This paper is written in the style of Cambridge International 9709 Paper 1 (Pure Mathematics 1). It is a practice paper, not an official examination paper.

List of formulae (as provided in the examination)

Quadratic equation: for $ax^2 + bx + c = 0$, $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion: $(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + b^n$, where $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

Arithmetic series: $u_n = a + (n-1)d$, $S_n = \frac{1}{2}n(a + l) = \frac{1}{2}n\{2a + (n-1)d\}$

Geometric series: $u_n = ar^{n-1}$, $S_n = \frac{a(1-r^n)}{1-r}$ ($r \neq 1$), $S_\infty = \frac{a}{1-r}$ ($|r| < 1$)

Circular measure: arc length = $r\theta$, sector area = $\frac{1}{2}r^2\theta$ (θ in radians)

Trigonometry: $\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$, $\sin^2 \theta + \cos^2 \theta \equiv 1$

Differentiation and integration: $\frac{d}{dx}(x^n) = nx^{n-1}$, $\int x^n dx = \frac{x^{n+1}}{n+1} + c$ ($n \neq -1$)

1 Solve the simultaneous equations

$$x + 2y = 5, \quad x^2 + y^2 = 10.$$

[3]

2

(a) Find the coefficient of x^3 in the expansion of $(1 + 3x)^5$. [2]

(b) Hence find the coefficient of x^3 in the expansion of $(2 - x)(1 + 3x)^5$. [2]

3 A sector of a circle has radius 9 cm and perimeter 30 cm.

(a) Find the angle of the sector in radians. [2]

(b) Find the area of the sector. [2]

4 In an arithmetic progression, the 4th term is three times the 1st term, and the sum of the first 6 terms is 96.

(a) Find the first term and the common difference. [3]

(b) Find the value of n for which the n th term is 106. [2]

5 The function f is defined by $f(x) = 4 - \frac{3}{x}$ for $x > 0$.

(a) State the range of f . [2]

(b) Find an expression for $f^{-1}(x)$ and state the domain of f^{-1} . [3]

(c) The function g is defined by $g(x) = 2x$ for $x > 0$. Solve the equation $fg(x) = 1$. [1]

6 The line $x + 2y = 10$ meets the x -axis at A and the y -axis at B .

(a) Write down the coordinates of A and B . [2]

(b) The point C lies on AB and OC is perpendicular to AB , where O is the origin. Find the coordinates of C and the exact length of OC . [4]

7

(a) Solve the equation $3 \sin \theta = 2 \cos^2 \theta$ for $0^\circ \leq \theta \leq 360^\circ$. [3]

(b) Solve the equation $\sin(2x + 40^\circ) = -0.5$ for $0^\circ \leq x \leq 180^\circ$. [4]

8 The equation of a curve is $y = 2x^3 - 9x^2 + 12x - 3$.

(a) Find the coordinates of the stationary points of the curve and determine the nature of each. [4]

(b) Find the set of values of x for which y is an increasing function of x . [2]

(c) Find the equation of the normal to the curve at the point where $x = 0$, giving your answer in the form $ax + by + c = 0$. [2]

9

(a) A curve is such that $\frac{dy}{dx} = 2 - \frac{8}{x^3}$ and the curve passes through the point (2, 5). Find the equation of the curve. [4]

(b) Find the area of the region enclosed by the curve $y = x^2 - 4x + 3$ and the x -axis. [3]

(c) Evaluate $\int_0^2 \frac{1}{(2x+1)^2} dx$. [2]

10

(a) Prove the identity $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} \equiv \frac{2}{\sin^2\theta}$. [3]

(b) Hence solve the equation $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 8$ for $0^\circ < \theta < 180^\circ$. [3]

(c) Given that $\sin x = \frac{2}{3}$ and that x is obtuse, find the exact value of $\tan x$. [2]

11

(a) The second term of a geometric progression is 6 and the sum to infinity is 27. Find the two possible values of the common ratio and the corresponding first terms. [4]

(b) The sum of the first n terms of a progression is given by $S_n = 2n^2 + 3n$.

(i) Find an expression for the n th term and show that the progression is arithmetic. [3]

(ii) Find the sum of the terms from the 11th term to the 20th term inclusive. [2]

12 A circle has equation $x^2 + y^2 + 4x - 2y - 20 = 0$.

(a) Find the coordinates of the centre and the radius of the circle. [2]

(b) Show that the point (1, 5) lies on the circle and find the equation of the tangent to the circle at this point, giving your answer in the form $ax + by = c$. [4]

END OF PAPER (Total: 75 marks)